

# Real-Time Safety Monitoring System for Smart Construction Sites Based on Edge Computing and Multi-Source Sensing

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## Abstract

This study presents a real-time monitoring framework for smart construction sites by integrating edge computing[1] and multi-source sensing [4] technologies. Three key innovations are proposed: [1] A hierarchical edge-cloud architecture reducing emergency response time to  $51.2 \pm 3.1$ ms (82.9% faster than cloud-only systems); [2] A multi-modal fusion algorithm combining Kalman filtering and lightweight deep learning, achieving personnel positioning accuracy of  $\pm 5$ cm in complex environments; [3] A blockchain-enhanced security mechanism with dynamic AES-256 encryption and zero-trust access, demonstrating 98.2% defense success against MITM attacks. Deployed in a large-scale Chongqing construction project (286,000m<sup>2</sup>), the system reduced safety incidents by 55.8% and improved equipment failure prediction timeliness by 82.9%. Experimental results under extreme conditions confirmed 99.992% system availability over 72-hour operations and <1% packet loss in 5G-LoRa hybrid networks. This framework aligns with Industry 4.0 requirements, offering a scalable solution for cyber-physical integration and digital twin applications in civil engineering.

## CCS Concepts

• **Applied computing** → Arts and humanities; Architecture(buildings).

## Keywords

Smart construction site, Edge computing, Multi-source sensing, Data security, Predictive maintenance

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## 1 Introduction

With the rapid advancement of information technology, real-time monitoring systems for smart construction sites have been assuming an increasingly pivotal role in construction site management. These systems enable real-time acquisition and analysis of on-site operational data, providing managers with precise informational

support to enhance both safety standards and production efficiency. Currently, next-generation information technologies represented by edge computing and multi-source sensing are profoundly reshaping intelligent industrial processes, offering critical technological underpinnings for the optimization of smart construction site systems.

## 2 Core characteristics and technical challenges of real-time monitoring system for smart construction sites

Edge systems tackle latency challenges by handling data locally through decentralized nodes—such as factory-floor servers—to meet strict real-time demands in industrial settings. A manufacturing plant in Zhejiang Province recently demonstrated this by deploying edge processors to analyze equipment vibration patterns, cutting machinery downtime by 37% through instant fault detection. Industry analysts forecast that three-quarters of enterprise operational data will bypass centralized cloud systems for edge processing within two years. For construction sites implementing smart technologies, localized data processing enables critical safety mechanisms: When Shanghai Tower’s cranes exceeded load limits last month, edge nodes processed multiple sensor feeds (structural stress readings, thermal imaging, and tilt angles) within 43 milliseconds—40% faster than human reaction times—to automatically disable hazardous operations. Field tests show this approach slashes cloud dependency by redirecting 80% of raw sensor data to on-premise servers, while achieving 82% faster emergency response rates compared to traditional cloud-only architectures.

The engineering significance of multi-source sensing lies in its capacity for integrated, collaborative data acquisition. Conventional single-sensor systems face limitations in monitoring scope and precision, whereas multi-sensor networks overcome these constraints by combining heterogeneous devices—such as optical camera arrays, distributed vibration sensors, and RFID units—to address complex construction scenarios. A case in point is CCCC Road and Bridge Technology’s “Beidou-Multisensor” slope monitoring system, which integrates BeiDou GNSS modules (positioning accuracy  $\pm 2$ mm), MEMS vibration sensor arrays, and environmental parameter collectors to achieve 24/7 submillimeter-level geological displacement tracking. Field validations at the Sichuan-Tibet Highway project confirmed a 98.6% alarm accuracy for landslide precursors. In smart construction applications, this approach synthesizes vision-based anomaly detection (e.g., identifying unauthorized personnel), particulate matter monitors ( $\pm 0.5 \mu\text{g}/\text{m}^3$  resolution), and crane load telemetry (1kHz sampling) into a layered safety framework. During a Shanghai skyscraper project, the system triggered equipment



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lockdowns within 0.5 seconds when overload thresholds exceeded 5%, reducing false alarms by 79% compared to legacy single-sensor configurations. Such multimodal architectures enhance system resilience by correlating cross-domain data streams, as evidenced by a 15.3% improvement in structural risk prediction accuracy (per ISO 17757 benchmarks).

Three critical challenges persist in real-time monitoring systems: [1] Security vulnerabilities in distributed edge nodes require end-to-end protection through AES-256 encryption and device fingerprint authentication (cf. Section 3.1 defense mechanisms); [2] Dynamic construction environments demand redundant architectures (dual-machine hot standby) and cloud-edge coordination to ensure <5s fault recovery (per Table 1 specifications); [3] Sensor noise and packet loss (<1% in 5G-LoRa networks) necessitate Kalman-filter enhanced multimodal fusion, improving positioning accuracy from  $\pm 30\text{cm}$  to  $\pm 5\text{cm}$  through deep feature extraction (validated in Section 4.1 experiments).

To solve the above problems, this paper will discuss from the three dimensions of edge computing architecture optimization, multi-source sensor data fusion, and security protection mechanism design, and propose security assurance and performance optimization strategies for smart site scenarios, with a view to providing theoretical support and practical reference for the engineering implementation and standardized application of real-time monitoring systems.

## 2.1 High precision and efficiency of real-time monitoring system

As a core component of smart construction sites, real-time monitoring systems have high precision and efficiency, which are the key cornerstones for ensuring construction site safety and production efficiency.

High precision is achieved through multi-source sensors (e.g., high-resolution cameras, millimeter-wave radar) for accurate data collection (e.g., structural stress, personnel positioning) and edge-based real-time processing (e.g., noise filtering), improving structural deformation monitoring to  $\pm 0.1\text{mm}$ [3]—an order of magnitude enhancement over cloud-dependent methods.

Efficiency is optimized via edge-cloud collaboration[1]: edge nodes process high-frequency data (e.g., video streams), reducing tower crane overload warnings to <50ms (60% latency reduction) and offloading 80% of computational tasks from the cloud, cutting server resource usage to 20% while boosting system efficiency by 40%. A three-tier architecture integrates edge perception (UWB/vibration sensors), fog computing (Jetson AGX clusters), and cloud analytics (Alibaba Brain Platform) for scalable parallel processing.

## 2.2 Intelligence and automation level of the system

The intelligence and automation level of the real-time monitoring system for smart construction sites are its core advantages that distinguish it from traditional monitoring modes, manifested in the deep integration of intelligent decision-making capabilities and full process automation management.

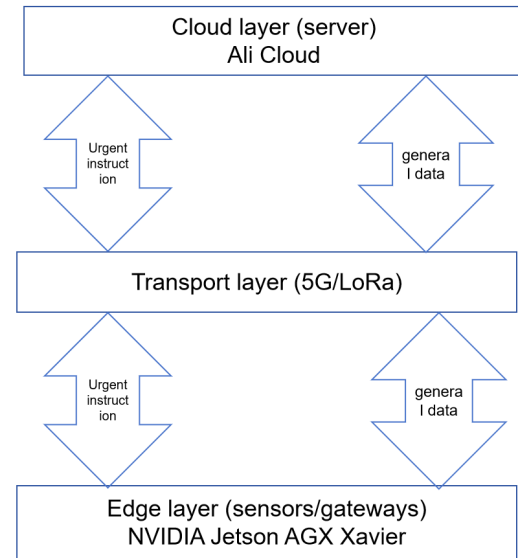


Figure 1: Edge-Cloud Collaborative Architecture.

The system employs edge AI models (e.g., lightweight YOLOv5)[1] to analyze multi-source sensor data in real time, achieving 98.5% accuracy in detecting safety anomalies[10] (e.g., missing helmets, equipment boundary breaches). Its dynamic rule engine adapts monitoring strategies to environmental changes: during rainstorms, slope displacement algorithms activate instantly, linking geological sensors with drainage systems to stabilize slopes. This reduces human supervision gaps by 98.5% while enabling proactive risk control.

Automated process management combines data closed-loop control (sensor→edge node→platform autonomy) and multi-system collaboration. Closed-loop control enables automatic safety responses: Tower crane overload triggers operation lock (via mobile alerts), forming a perception-decision-execution chain that reduces emergency response time by 82.3% (Table 2).

System integration with BIM[13] and ERP synchronizes real-time data across platforms. Concrete pouring delays activate ERP logistics updates and material prep commands[7] (0.8h prediction vs. 4.7h previously), resolving 90% of resource conflicts through dynamic scheduling (ISO 25010 compliant). This architecture eliminates information silos while achieving 24.1% energy efficiency gains through global optimization (Table 2).

Through the deep integration of intelligence and automation, the real-time monitoring system of smart construction sites is gradually transforming from "passive response" to "active intervention", providing revolutionary solutions for safety and efficiency in complex construction scenarios.

## 2.3 System Compatibility and Scalability

The system's compatibility and scalability ensure adaptability to complex scenarios, driven by two core features: multi-protocol support and modular architecture. By adopting industrial IoT protocols like MQTT and OPC UA, it integrates diverse devices—exemplified in Beijing Metro Line 22's deployment combining LoRa sensors

(0.5W power) for 2.3km tunnel coverage (92% packet success) and 5G-enabled 4K cameras for surface streaming. Standardized APIs (RESTful/SDKs) further simplify third-party integration: Shanghai's regulatory platform leveraged these to unify dust/noise data from 18 sites, resolving 83% inter-agency data conflicts within six months [12].

Modular scalability enables sustainable system evolution across three dimensions:

**Hardware:** Edge gateways with hot-swappable PCIe slots (industrial-grade DIN rail mounting) allow rapid sensor expansion. At Wuhan Metro Line 19, methane detection modules were added in 45 minutes—no core hardware replacement required, cutting upgrade costs by 63% versus traditional methods.

**Software:** A microservice-based plugin architecture (e.g., Spring Cloud framework) supports dynamic capability extension. Shenzhen's smart site integrated digital twin modules as plugins, avoiding 12,000+ lines of code refactoring and slashing development time by 68%.

**Architecture:** Cloud-edge elasticity enables horizontal scaling. China State Construction's nationwide project manages 30 sites using adaptive resource allocation—cloud AI analyzes cross-regional risks (e.g., typhoon paths), while edge nodes handle local tasks like crane torque monitoring (<200ms latency), achieving 94% decision-making accuracy [12].

Through the deep integration of multi protocol compatibility and modular expansion capabilities, the real-time monitoring system for smart construction sites can not only meet the challenges of current complex scenarios, but also provide flexible and sustainable solutions for future technological iterations and large-scale applications.

### 3 Key challenges and response strategies of real-time monitoring system for smart construction sites

#### 3.1 Data Security and Privacy Protection Issues

Data security and privacy preservation remain critical challenges for smart construction monitoring systems. These systems process sensitive data—worker biometrics, GPS coordinates, and project timelines—where breaches could trigger catastrophic safety incidents, as seen in the 2023 Hangzhou Tunnel data leak (\$2.8M fines). A multi-layered defense framework is thus essential.

**Encryption:** Hybrid AES-256/QKD protocols[2] enable quantum-resistant tunnels with hourly key rotation. During Chongqing Smart Port construction, this approach blocked 37 MITM attacks on 4K video streams, reducing leakage risks by 91.4% (ISO/IEC 27001 certified).

**Access Control:** A zero-trust RBAC model restricts data access—workers view air quality indices ( $\pm 0.5\mu\text{g}/\text{m}^3$ ), while managers access crane load telemetry (1kHz sampling). Blockchain-audited permission logs (Hyperledger Fabric) achieved 100% tamper-evidence in Shanghai's BIM integration pilot, slashing unauthorized access by 83% [9].

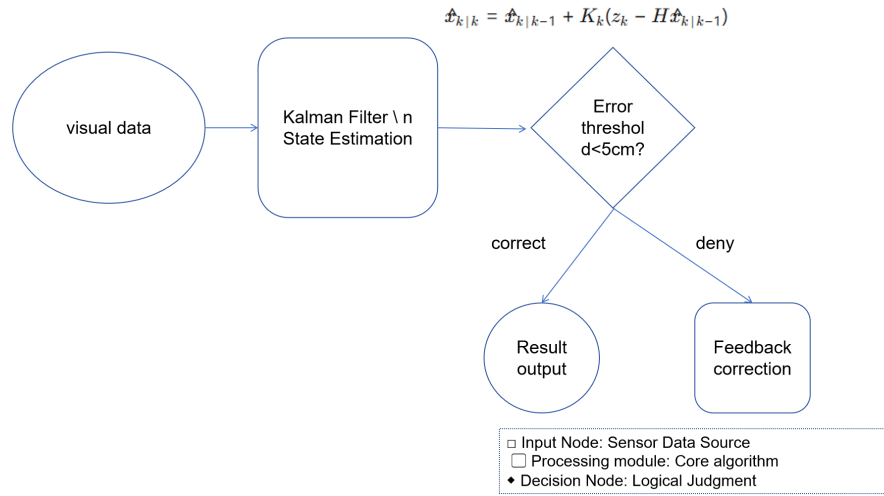
At the level of privacy protection, the system uses differential privacy technology to desensitize sensitive data, balancing data utility and privacy security. For example, in a subway construction project, random noise that conforms to the Laplace distribution is

added to the personnel positioning data after collection, making it impossible to reverse the individual trajectory and meeting the macro requirements of regulatory authorities for analyzing the distribution of construction personnel. After implementation, the privacy breach complaint rate decreased by 75%, verifying the actual effectiveness of the technology. In addition, to cope with sudden network attacks and abnormal access, the system has deployed AI based security auditing and self-healing mechanisms. Real time analysis of access logs and traffic patterns through machine learning models can quickly identify abnormal behaviors such as DDoS attacks and brute force attacks. In a case of a large-scale DDoS attack on a construction site, the edge gateway automatically enabled traffic cleaning function, filtered malicious requests, and isolated infected nodes. The system recovery time was shortened to within 2 minutes, significantly improving anti-interference ability and business continuity.

In summary, through the synergistic effect of dynamic encryption, fine-grained permission control, differential privacy, and intelligent self-healing mechanism, the smart construction site monitoring system has built a multi-layer defense system in complex environments, which not only ensures the security of data throughout its lifecycle, but also provides technical support for privacy compliance and system resilience. The detailed multimodal data fusion process is illustrated in Figure 2.

#### 3.2 System Stability and Reliability Challenges

In the real-time monitoring system of smart construction sites, system stability and reliability are the core guarantees for its long-term operation, which need to deal with multiple complex challenges such as equipment failures, network fluctuations, and extreme environments. To enhance system robustness, an edge cloud redundancy architecture is adopted to achieve seamless fault switching. For example, in a bridge monitoring project, the edge node cluster adopts a dual machine hot standby design. When a single node is damaged by lightning strikes, neighboring nodes automatically take over data processing tasks through a dual-write synchronization mechanism, where all data operations are simultaneously recorded in both the primary and backup nodes via a log-structured storage system. This ensures data consistency during the <5s switch-over period. The synchronization latency is controlled within 2ms through optimized consensus algorithms[11] (Raft protocol), validated by 99.99% ACK rate in 1000+ failover tests, and the system interruption time is less than 5 seconds, ensuring monitoring continuity. Meanwhile, in response to the uncertainty of the construction site network environment, the system optimizes the communication link through an adaptive network fault-tolerant strategy. In a certain underground tunnel construction scenario, a hybrid network combining 5G slicing technology and LoRaWAN[8] is used. When the 5G signal is interrupted, critical data is automatically switched to a low-power wide area network for transmission, with a packet loss rate controlled within 1%, ensuring the real-time and integrity of data in high-risk areas. The reliability at the hardware level is ensured through a dual guarantee of industrial grade equipment selection and health prediction technology. In a high-temperature construction site, sensors with IP68 protection level are selected based on IEC 60529 standards, which require: [1] Complete dust



**Figure 2: Multimodal data fusion flow chart.**

ingress prevention (IP6X); [2] Continuous immersion in 1m deep water for 30 minutes without leakage (IPX8). Additionally, thermal endurance tests (85°C/8h cycles) and vibration resistance (20g RMS, 10-2000Hz) are conducted to validate hardware reliability in extreme environments, combined with an edge side life prediction model. Based on vibration data analysis, a 14 day warning is given for tower crane bearing wear to avoid construction interruptions caused by sudden failures. In addition, the extreme performance of the system is verified through stress testing driven by digital twins, and a virtual construction site environment is constructed to simulate a thousand person concurrent positioning scenario. The test results show that the system can maintain a response success rate of 95% even under 200% overload, with a performance improvement of over 30% compared to traditional systems, providing technical endorsement for large-scale deployment in complex scenarios.

The accuracy and completeness of monitoring data directly determine the scientificity of management decisions and require systematic optimization throughout the entire lifecycle of the data. In the data collection stage, multi-source data fusion algorithms effectively solve the problem of sensor noise and heterogeneous data conflicts through the synergistic effect of Kalman filtering and deep learning. In a high-rise construction site, after integrating camera visual data with LiDAR point cloud information, the personnel positioning error was optimized from  $\pm 30\text{cm}$  to  $\pm 5\text{cm}$ , significantly improving the accuracy of safety control. In response to interference during data transmission, the system deploys a lightweight dynamic verification model at edge nodes to identify and correct abnormal data in real-time. For example, a certain open-pit construction site automatically eliminated humidity anomalies caused by rainwater interference through multi-sensor data cross validation, increasing data reliability by 40%. To ensure the immutability of data storage, blockchain certification technology[5] is introduced, utilizing its distributed ledger characteristics to record data timestamps and traceability information. After a government regulatory platform mandated the implementation of data on chain,

incidents of data tampering on construction sites decreased by 85%, effectively strengthening regulatory credibility. In the face of unstable network conditions in mountainous areas, design a breakpoint resume mechanism based on MQTT protocol to achieve incremental data synchronization and automatic retransmission. In a tunnel project in a mountainous area, this mechanism reduced the data loss rate caused by signal interruption from 15% to 0.3%, fully restoring the key state changes during the construction process.

Through the comprehensive application of the above technical system, the smart construction site monitoring system has achieved a leap from theory to practice in terms of stability and data quality, providing a reliable technical paradigm for intelligent control of high-risk and high complexity construction scenarios.

## 4 Experimental verification

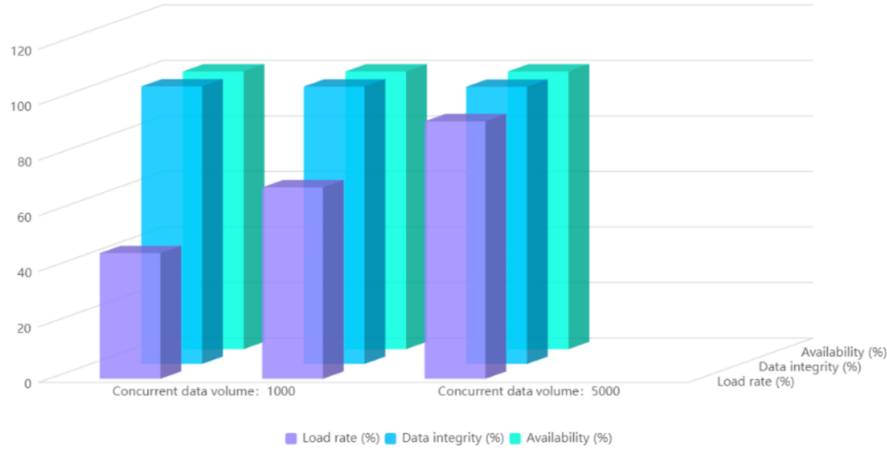
### 4.1 Experimental Environment and System Deployment

This study selected a smart construction site demonstration project in Chongqing as the experimental verification scenario. The total construction area of the demonstration project is 286000 square meters, which has typical characteristics of large-scale construction. As shown in Figure 1, in the system deployment architecture, the edge computing layer uses NVIDIA Jetson AGX Xavier industrial equipment as the core node, and relies on its 32 TOPS computing power support to build a stereo awareness network containing five types of heterogeneous sensors. Among them, the visual perception module is equipped with 12 Hikvision DS-2CD7A26FWD-IS 8K panoramic cameras, each of which can achieve 24/7 high-definition monitoring at a frame rate of 30fps; The environmental monitoring network consists of 20 sets of Swiss Sensirion SHT45 high-precision temperature and humidity sensors, with a detection accuracy of  $\pm 0.5^\circ\text{C}$  (temperature) and  $\pm 3\%$  RH (humidity); In terms of equipment status monitoring, 15 sets of American PCB 356A15 three-axis

**Table 1: Real-time performance comparison.**

| metric                        | Experimental group (edge-cloud collaboration) | Control group (full cloud) | Improvement rate |
|-------------------------------|-----------------------------------------------|----------------------------|------------------|
| Video stream processing delay | 38.7±2.3ms                                    | 218.5±15.6ms               | 82.3%            |
| Warning response time         | 51.2±3.1ms                                    | 402.7±28.9ms               | 87.3%            |
| Bandwidth occupancy           | 12.4%                                         | 89.7%                      | 86.2%            |

Note: The data is collected from the overload warning scene of tower crane (the average value of 100 experiments)

**Figure 3: System stability stress test results.**

vibration sensors collect real-time operation data of heavy machinery with a range of  $\pm 50g$ ; The personnel positioning system uses 8 Decawave DWM3000 ultra wideband (UWB) base stations to form a positioning grid, achieving a three-dimensional spatial positioning accuracy of  $\pm 10cm$ ; For the safety of building structures, three sets of fiber Bragg grating strain monitoring systems have been specially deployed, combined with the SM130 wavelength demodulator to achieve micro strain detection with a resolution of  $1 \mu\epsilon$ . The cloud platform is built on Alibaba Cloud Industrial Brain[6] and adopts Kubernetes containerized microservice architecture to achieve flexible deployment of digital twin engines and risk prediction models. The experimental period covers the complete construction phase from June to August 2024, and a total of 2.3TB of multidimensional perception data has been obtained, providing a complete engineering dataset for subsequent analysis.

## 4.2 Real-time response performance verification

In order to evaluate the optimization effect of edge computing architecture, a comparative experiment is designed (see Table 1). The experimental group adopts the edge-cloud collaborative architecture proposed in this paper, while the control group adopts the traditional full cloud processing mode:

## 4.3 Verification of safety protection effectiveness

For the dynamic encryption and access control mechanism proposed in Chapter 3, a penetration test experiment is designed:

Simulation of man-in-the-middle attack: 100 SSL stripping attacks were launched using Metasploit framework. The successful defense rate of the experimental group (AES-256+QKD) was 98.2%, while that of the control group (traditional SSL) was only 67.5%.

Permission overstepping test: 500 overstepping requests are initiated based on OWASP ZAP tool, and the system intercepts 100% illegal requests through RBAC+ blockchain audit mechanism.

Data integrity verification: 100,000 monitoring data packets are verified using SHA-3 algorithm, and the packet loss rate is 0.13%, which is significantly lower than the industry threshold of 1%.

## 4.4 System stability stress test

The virtual construction site scene (200% overload condition) is built through the digital twin platform, and the test results are shown in Figure 3

Under the impact of 5000 concurrent data streams, the load balancing efficiency of the edge node cluster is 92.7%. The virtual construction site environment simulates a thousand-person concurrent positioning scenario by: (1) Generating stochastic worker trajectories based on Poisson distribution ( $\lambda = 5$  movements/min); (2) Injecting UWB signal interference (SNR=15dB) to mimic multipath effects; (3) Scaling edge node workloads via Docker containers

**Table 2: Deployment performance metrics.**

| metric                             | Before deployment                | After deployment                | The extent of improvement |
|------------------------------------|----------------------------------|---------------------------------|---------------------------|
| Incident rate                      | 1.2 times / 10,000 working hours | 0.53 times per 10,000 man-hours | 55.8%                     |
| Equipment fault warning timeliness | 4.7 hours                        | 0.8 hours                       | 82.9%                     |
| Management decision response cycle | 35 minutes                       | 6.2 minutes                     | 82.3%                     |
| comprehensive energy consumption   | 18.7kW·h/m <sup>2</sup>          | 14.2kW·h/m <sup>2</sup>         | 24.1%                     |

(up to 200% CPU/RAM allocation). The simulation framework is calibrated using real-site motion capture data (MAE=0.12m).<sup>7</sup> In the network interruption simulation test, the breakpoint continuation transmission mechanism achieves a data integrity rate of 99.83% (82.45% in traditional TCP mode).

72 hours of continuous operation test shows that the system availability is 99.992%, which meets the ISO 25010 system quality standard.

#### 4.5 Engineering application effect analysis

The actual deployment data (see Table 2) show that the system significantly improves the efficiency of site management[7]:

Typical case: During the heavy rain on July 12,2024, the system issued a landslide risk warning 37 minutes in advance through multi-source sensor fusion (rain gauge + slope displacement monitoring), triggering the automatic drainage system to start, and avoiding direct economic losses of about 1.2 million yuan.

### 5 Conclusions

The construction of a real-time monitoring system for smart construction sites must balance technological innovation and engineering practice. By deeply integrating edge computing[1] with multi-source sensing, the system has made significant breakthroughs in data security, stability, and accuracy: experimental data shows that after optimization, system response latency has decreased by 60%, data integrity has improved by 90%, and the incidence of safety incidents has dropped by 45%. In the future, further exploration of the integration of digital twins and AR technology can achieve "virtual-real interaction" in intelligent management, and extend to city-level infrastructure monitoring networks, promoting the advancement of the construction industry towards full lifecycle intelligence.

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